

McGill
CHEM 120

Summer 2025, Chapter 5 Notes



Table of Contents

Chapter 5. Thermodynamics II (Ch 12)

5.1. Entropy

5.1.1. 2nd Law of Thermodynamics (Entropy)

5.1.2. Factors Affecting Entropy Changes

5.1.3. Example

5.1.4. Practice

5.1.5. Practice

5.2. 3rd Law of Thermodynamics

5.2.1. 3rd Law of Thermodynamics

5.2.2. Summary of the Laws of Thermodynamics

5.3. Gibbs Free Energy + Spontaneity

5.3.1. Gibbs Free Energy and Spontaneity

5.3.2. Spontaneity

5.3.3. Example: Spontaneity of a Reaction

5.3.4. Predicting Spontaneity Based off Entropy

5.4. Connecting Gibbs Free Energy and Equilibrium

5.4.1. Chemical Equilibrium and ΔG Equations

5.4.2. ΔG and K

5.4.3. Example

5.4.4. Chemical Equilibrium and ΔG Practice

5.4.5. Chemical Equilibrium and ΔG Practice #2

5.4.6. Chemical Equilibrium and ΔG Practice #3

5.4.7. Van't Hoff Practice Problem

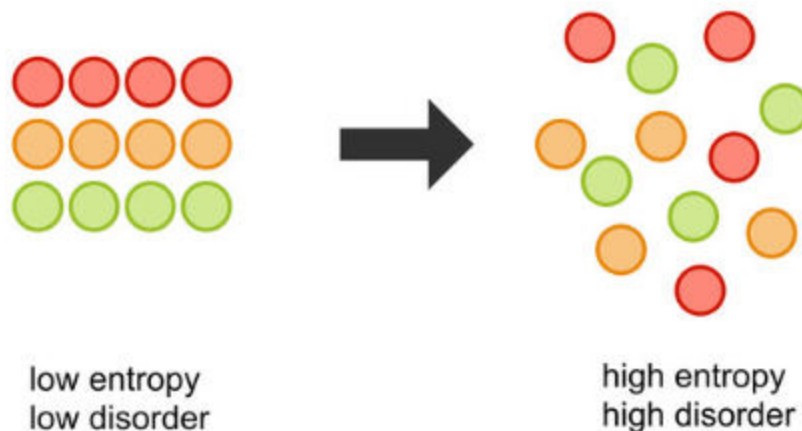
5. Thermodynamics II (Ch 12)

5.1 Entropy

5.1.1

2nd Law of Thermodynamics

Entropy (S) is a measure of disorder or randomness in a system.



The **2nd Law of Thermodynamics** states that disorder of the universe increases in a spontaneous process.

In other words, things naturally tend to go towards disorder and the universe is constantly getting more and more disordered!



Entropy Equations

Entropy is defined as:

$$\Delta S_{surr} = \frac{q_{surr}}{T}$$

ΔS_{surr} =entropy of the surroundings (in J/molK)

q_{surr} =heat in the surroundings (in J)

T =Temperature (in K)

Recall:

$$q_{surr} = -q_{sys}$$

At constant pressure and temperature:

$$\Delta H_{sys} = q_{sys}$$

$$\Delta S_{surr} = -\frac{\Delta H_{sys}}{T}$$

The **2nd law of thermodynamics** states that whenever a spontaneous event takes place in the universe, the total entropy of the universe increases:

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surroundings} > 0$$

Chemical Reactions

In the same way we can use standard heats of formation to calculate a reaction enthalpy at room temperature and

pressure, we can use S° (values will be provided to you in a table)

$$\Delta S^\circ = [(\sum nS^\circ_{products}) - (\sum nS^\circ_{reactants})]$$

Factoring Affecting Entropy Changes

Entropy for Different Phases

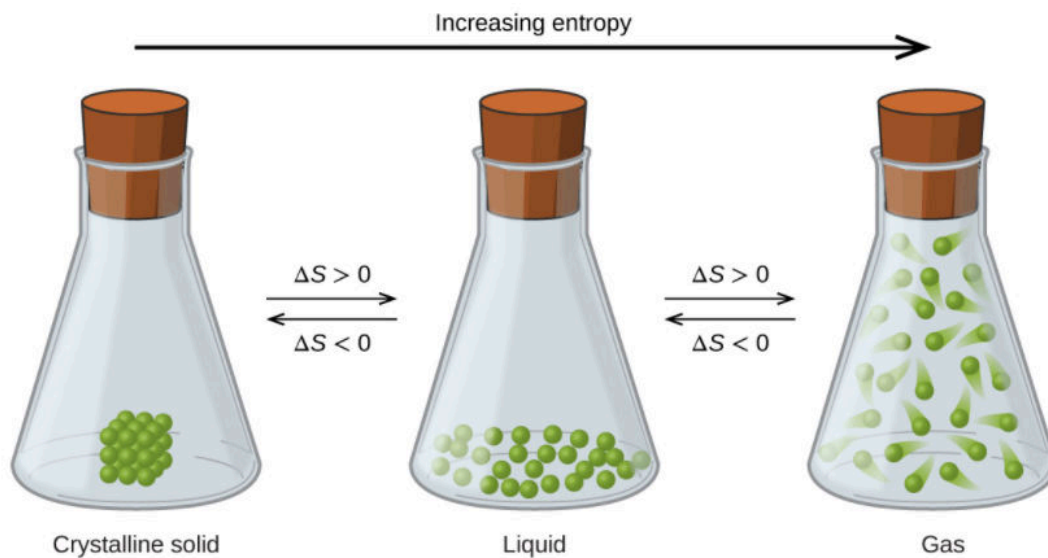
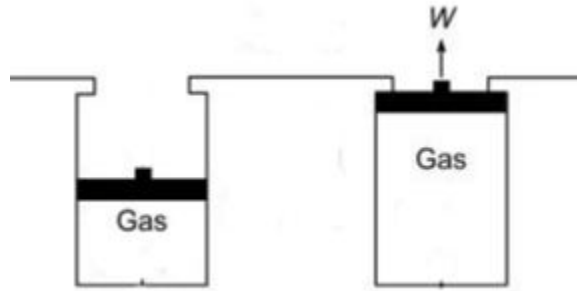


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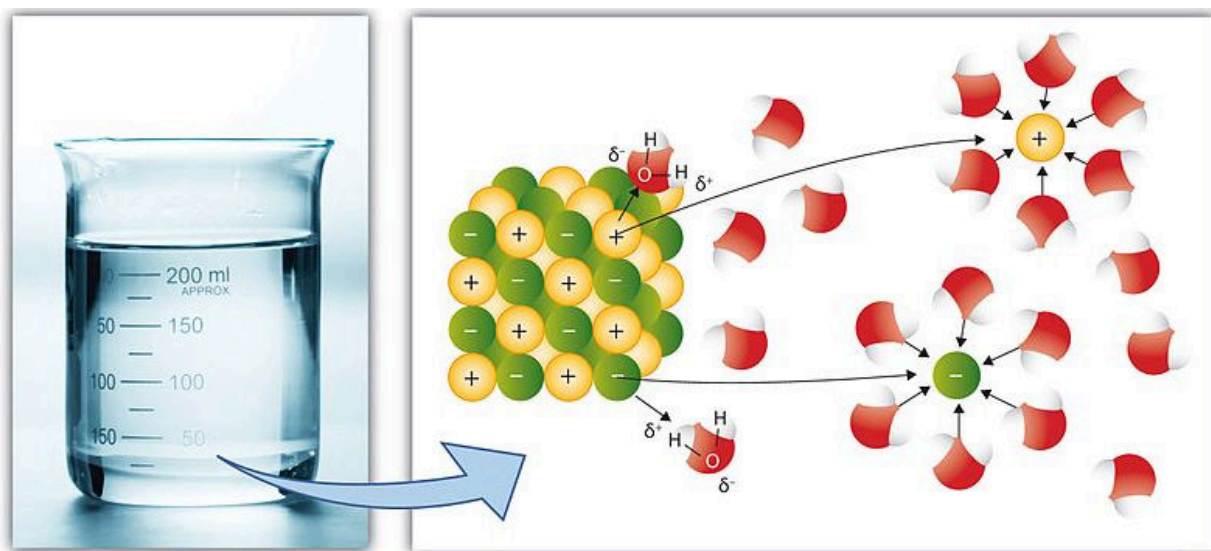
Entropy When a Higher Pressure is Applied to a Gas

If we have **two gases**, and **apply a higher pressure to one** of them, will the gas with the higher pressure applied have more or less entropy?



The gas with the higher pressure applied to it will have (more/less) _____ organized molecules and therefore there is (more/less) _____ disorder, meaning there is (more/less) _____ entropy!

Entropy of a Solid Alone vs Dissolved in a Liquid



Example:

KCl(s) vs KCl(aq)

There is (more/less) _____ entropy when the solid is dissolved in a solvent.

Entropy of a Gas Alone vs Dissolved in a Solvent



Example: Think of shaking a bottle of pop $\text{CO}_2(\text{g})$ vs $\text{CO}_2(\text{aq})$

The gas would have (more/less) _____ more entropy when dissolved in the solvent.

Entropy of Similar Substances in the Same Physical State

How would entropy change with increasing mass?

Entropy would (increase/decrease) _____

Example:

$\text{H}_2(\text{g})$ (MM=2g/mol) vs $\text{Ar}(\text{g})$ (MM=40g/mol) vs $\text{CO}_2(\text{g})$ (MM=44g/mol)

How would entropy change as we increase the number of atoms in the molecule?

Entropy would (increase/decrease) _____

Example:

$\text{CH}_4(\text{g})$ vs $\text{C}_2\text{H}_6(\text{g})$ vs $\text{C}_{10}\text{H}_{22}(\text{g})$

How would entropy be different if we compared the following two molecules?

Cyclopentane vs C_5H_{12}

Would entropy increase or decrease with increasing molecular freedom or flexibility? _____

Entropy in a Chemical Equation

Example:

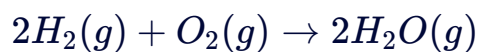


If we were asked if entropy increases or decreases in this equation, what would you consider?

Therefore, entropy _____ We could also say that ΔS is _____ 0

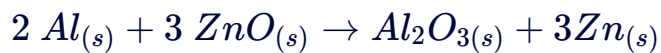
Example: Qualitatively Estimating Entropy Changes

Determine if the entropy change will be positive or negative for the following reactions:



5.1.4

Using the following data, calculate the standard entropy of the reaction shown below. Enter your answer in units of J / (mol*K)



$$S^\circ(\text{Al}_{(s)}) = 28.3 \text{ J mol}^{-1} \text{K}^{-1}$$

$$S^\circ(\text{Zn}_{(s)}) = 41.6 \text{ J mol}^{-1} \text{K}^{-1}$$

$$S^\circ(\text{ZnO}_{(s)}) = 43.9 \text{ J mol}^{-1} \text{K}^{-1}$$

$$S^\circ(\text{Al}_2\text{O}_{3(s)}) = 51.0 \text{ J mol}^{-1} \text{K}^{-1}$$

Answer

5.1.5

Using the ΔH^0 of fusion for water 6.03 kJ/mol and the ΔS^0 of fusion for water 22.1 J K⁻¹ mol⁻¹, calculate the ΔS_{univ} for ice melting at -10°C, 0°C and 10°C. Remember: the universe transfers heat in a reversible way.

A ΔS_{univ} at -10C

B ΔS_{univ} at 0C

C ΔS_{univ} at 10C

5.2 3rd Law of Thermodynamics

5.2.1

3rd Law of Thermodynamics

- The Third Law states: All perfect crystals in equilibrium at 0K have the same entropy, defined as 0
- Conceptually this makes sense given our understanding of order and disorder. The system described in the third law would be the most ordered system imaginable.

No Vibrational Motion at 0K

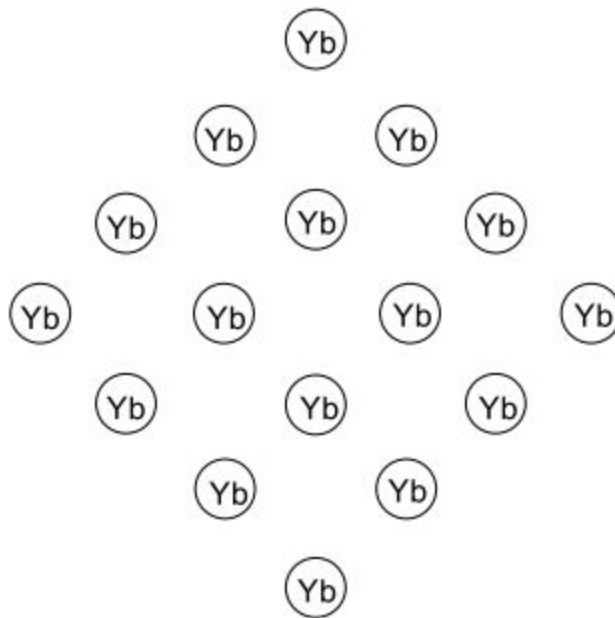


Figure 2.2.4

Summary of the Laws of Thermodynamics

WIZE CONCEPT

It is more important to understand each law and the related equations than to memorize what each law is!

0th Law: 2 bodies that are in thermal equilibrium with a 3rd body are in thermal equilibrium with each other as well! All 3 bodies would be at the same temperature.

1st Law: Energy is conserved and cannot be created and destroyed. Energy can only be transformed into different forms like heat.

$$\Delta U_{surr} = -\Delta U_{sys}$$

$$\Delta U = q + w \dots \text{where } w = -P\Delta V$$

2nd Law: Entropy (disorder) in the universe will spontaneously increase. Energy like to be more spread out!

$$\Delta S^{o}rxn = \left[\left(\sum n \Delta S^{o}_{products} \right) - \left(\sum n \Delta S^{o}_{reactants} \right) \right]$$

$$\Delta G = \Delta H - T\Delta S$$

3rd Law: A perfect crystal has 0 entropy at a temperature of 0K (hypothetical temperature that is the absolute lowest temperature). Any temperature higher than this would have entropy and there is more entropy as temperature increases!

5.3 Gibbs Free Energy + Spontaneity

5.3.1

Gibbs Free Energy and Spontaneity

- It is not always convenient to calculate the entropy of the system and the surroundings to determine whether a process is spontaneous
- Gibbs showed that at constant T and p another state function, ΔG , of the system can predict whether a reaction is spontaneous or not.
 - under constant T:

$$\Delta G = \Delta H - T\Delta S$$

- under constant T and P:
- If $\Delta G > 0$ the reaction is _____ (_____ in the reverse direction)
- If $\Delta G < 0$ the reaction is _____ and reversible/irreversible _____
- Just like enthalpy and entropy Gibbs free energy for a reaction can be calculated

$$\Delta G_{products} - \Delta G_{reactants} = \Delta G_{rxn}$$

Note: we can use the same equation we saw for enthalpy of the reaction and entropy of the reaction. For Gibbs Free Energy just be aware that if the tables are given with values at 25C, this equation can only be used when reaction conditions are at 25C!

- Gibbs free energy is a state function which allows us to set up the following relation

$$\Delta G_{forward} = -\Delta G_{reverse}$$

- Also recall how for ΔH°_f of aqueous ions, it was not possible to separate the cation and anion contributions
 - $\Delta H^\circ_f = 0$ for H^+
 - we can do the same for entropy and Gibbs free energy: $\Delta S^\circ = 0$ for H^+ and $\Delta G^\circ_f = 0$ for H^+

Spontaneity

Look at each row in the following table, use $\Delta G = \Delta H - T\Delta S$

Which are spontaneous, which are not?

Which are spontaneous only at a high/low temperature?

If we change the temperature, which could become non-spontaneous??

Which one is entropy driven to be spontaneous and which one is enthalpy driven to be spontaneous?

ΔG	ΔH	ΔS
-	-	+
-	-	-
-	+	+
+	+	-

We can be even more specific with our temperatures, rather than saying high or low temperatures to be spontaneous:

Write a < or > sign:

For Row 2: $T \frac{\Delta H}{\Delta S}$

For Row 3: $T \frac{\Delta H}{\Delta S}$

Now what if for row 2 we had placed the opposite sign?

What about for row 3 if we placed the opposite sign?

5.3.3

Using the following table, predict which of the following reactions would be spontaneous at 25°C:

	ΔH	ΔS
Reaction 1:	$9.25 \frac{kJ}{mol}$	$25.1 \frac{J}{molK}$
Reaction 2:	$2.85 \frac{kJ}{mol}$	$-125 \frac{J}{molK}$

If either of the reactions is non-spontaneous, at what temperature might it become spontaneous?

Consider reaction 1:

Now consider reaction 2:

Predicting Spontaneity from Entropy

- There are two main ways you may be asked to figure out the spontaneity of a process
 - If you are given the change in entropy, temperature, AND change in enthalpy (ΔH)
 - If you are given the change in entropy of the system AND the change in entropy of the surroundings (from the 2nd law of thermodynamics)
 - Remember $\Delta S_{universe}$ allows us to predict spontaneity and $\Delta S_{universe} = \Delta S_{system} + \Delta S_{surr}$

EXAM TIP

This chart can be used to predict the spontaneity of a process if you're only given the change in entropy of the system and surroundings

ΔS_{system}	$\Delta S_{surroundings}$	Is it spontaneous?
+	+	spontaneous
-	-	non-spontaneous
-	+	depends on the magnitude of each
+	-	depends on the magnitude of each
$\Delta S_{universe} = 0$		Equilibrium

5.4 Connecting Gibbs Free Energy and Equilibrium

5.4.1

Chemical Equilibrium and ΔG

- In general the Gibbs free energy can be related to the reaction quotient by the following expression

$$\Delta G = \Delta G^o + RT \ln Q$$

Recall:

$\Delta G < 0$ for a spontaneous process

$\Delta G = 0$ at equilibrium

- If the reaction is at equilibrium then:

$$\Delta G = 0 \quad Q = K$$

$\therefore$

$$\Delta G^o = -RT \ln K$$

We can rearrange this equation to solve for K:

$$\ln K = \frac{\Delta G^o}{-RT}$$

$$e^{\ln K} = e^{\left(\frac{\Delta G^o}{-RT}\right)}$$

$$\ln e(K) = e^{\left(\frac{\Delta G^o}{-RT}\right)}$$

$$K = e^{\left(\frac{\Delta G^o}{-RT}\right)}$$

i WIZE TIP

We just covered 3 important equations to know for your exam!

1)

$$\Delta G = \Delta G^o + RT \ln Q$$

2) Is what we get when we let $\Delta G=0$ in the above equation:

$$\Delta G^o = -RT \ln K$$

3) We get when we solve for K using equation 2:

$$K = e^{\left(\frac{\Delta G^o}{-RT}\right)}$$

Note: the e in the above equation is a button on your calculator! We will look at some problems so you can get some practice getting familiar with that button!

Gibbs Free Energy and Chemical Reactions

Gibbs free energy is a state function, so just like enthalpy we can use Gibbs free energies of formation to determine the change in Gibbs free energy for a reaction. This equation can only be used when the temperature is 298K!

$$\Delta_r G^o = \sum_i v_i \Delta_f G_{m,i}^o(\text{products}) - \sum_j v_j \Delta_f G_{m,j}^o(\text{reactants})$$

van't Hoff Equation

$$\ln \left(\frac{K_2}{K_1} \right) = \frac{-\Delta H^o}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

****memorize this equation as well!**

ΔG° and K

If $\Delta G^\circ < 0$ then K is...

If $\Delta G^\circ > 0$ then K is...

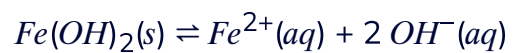
If $\Delta G^\circ = 0$ then K is...

Note: that ΔG° does not determine spontaneity (remember ΔG does!) ΔG° only tells us about the value of K

- only when $Q=1$ or everything is in standard state, or everything is in standard state, will ΔG° tell us about spontaneity

5.4.3

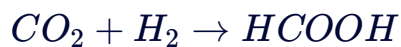
Consider the following reaction at 25°C:



Calculate ΔG° for the reaction. K_{sp} for $\text{Fe}(\text{OH})_2$ is 1.6×10^{-14} .

5.4.4

$\Delta_r S^\circ$ and $\Delta_r H^\circ$ are known to be $-225.3 \text{ J K}^{-1} \text{ mol}^{-1}$ and $-32.8 \text{ kJ mol}^{-1}$ respectively for the following reaction.



- Is this reaction spontaneous under standard conditions at $T = 298\text{K}$ and $P = 1 \text{ atm}$?
- Are reactants or products favoured in the above conditions?
- At what temperature does the reaction favour the opposite?

a.

b.

c.

5.4.5

The equilibrium constant for the decomposition of $\text{N}_2\text{O}_{4(g)}$ into $\text{NO}_{2(g)}$ has an equilibrium constant $K = 660021$ at 298K.

- a. Write the balanced chemical reaction for the decomposition of $\text{N}_2\text{O}_{4(g)}$.
- b. What is ΔG° for the above reaction?
- c. It is known that $\Delta S^\circ = +165 \text{ J mol}^{-1} \text{ K}^{-1}$ and $\Delta H^\circ = +15.9 \text{ kJ mol}^{-1}$. Assuming ΔH° and ΔS° are temperature independent, calculate the equilibrium constant, K .

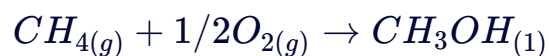
a.

b.

c. Solve for K :

5.4.6

In British Columbia there has been fierce debate over whether we should increase or halt production of LNG liquefied natural gas (CH_4). One alternative to liquefying CH_4 (bp = -161.5°C) is to convert it directly into methanol (CH_3OH) which is a liquid at room temperature and pressure in the reaction shown below.



Substance	$\Delta H^\circ_{f,m}$ (kJ mol $^{-1}$)	$S^\circ_{f,m}$ (J mol $^{-1}$)
$\text{CH}_{4(g)}$	-74.8	186
$\text{O}_{2(g)}$		205
$\text{CH}_3\text{OH}_{(l)}$	-238.7	126.8

- Calculate ΔH° , ΔS° and ΔG° at 25°C for the conversion of natural gas to methanol. Is this process spontaneous?
- Calculate the equilibrium constant for this reaction at 25°C .
- What are three things you could do to favour the production of methanol?
- Under the following conditions is the reaction spontaneous? (see below)

b.

c.

d.

$$T = 1000\text{K}$$

$$p(O_2) = 100 atm$$

$$p(CH_4) = 200 atm$$

$$p(CH_3OH) = 1 atm$$

5.4.7

Consider the following reaction



$$\Delta H^0 = -180 \text{ kJ/mol}$$

K is measured to be 9.9×10^2 at 800 K , what temperature would the reaction need to be run at to get an equilibrium constant of 1×10^6 ?
